

William H. Hayt, Jr. • Jack E. Kemmerly • Steven M. Durbin



Engineering Circuit Analysis

Eighth Edition

Chapter 18 Fourier analysis

Harmonics

Total response:

$$v(t) = v_n(t) + v_f(t)$$

Let us focus on **forced response** now:

Consider cosinusoidal forcing function

$$v_1(t) = V_m \cos \omega_0 t$$

ω_0 is called ***Fundamental Frequency***

$n\omega_0$ are called ***Harmonics*** with $n=1,2,3,\dots$

All harmonics add together and modify the final waveform.

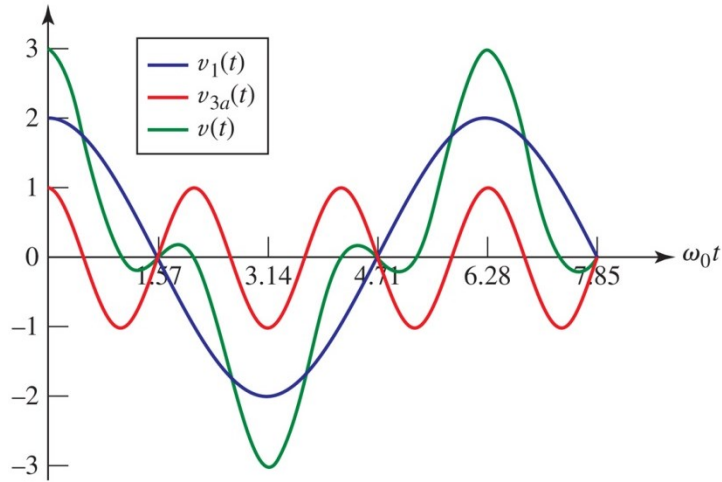
However, period of the resulting signal remains the same:

$$T = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

Harmonics

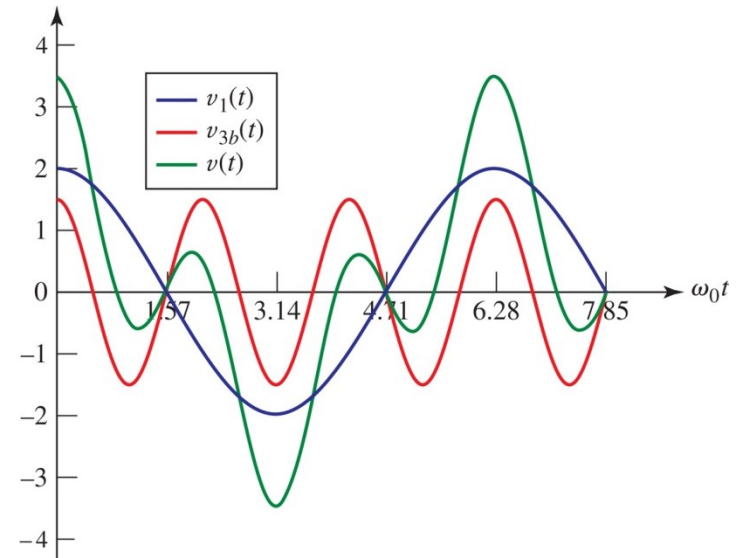
Example of 3rd harmonics, $n=3$ for forcing function $v_1(t) = 2 \cos \omega_0 t$

$$v_{3a}(t) = \cos 3\omega_0 t$$



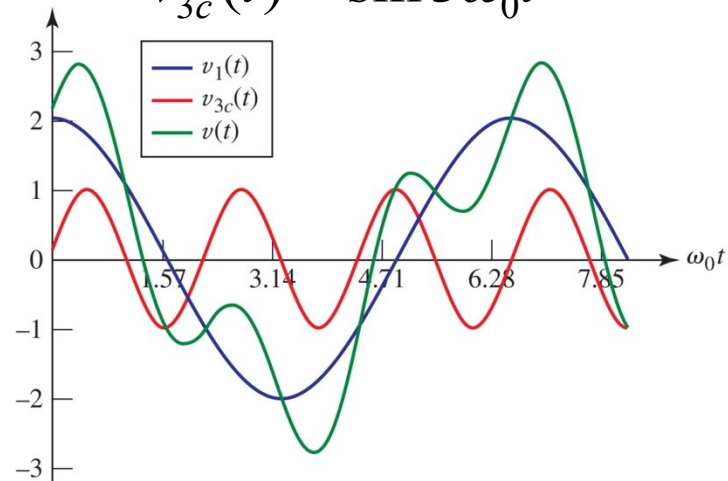
(a)

$$v_{3b}(t) = 1.5 \cos 3\omega_0 t$$



(b)

$$v_{3c}(t) = \sin 3\omega_0 t$$



(c)

The Fourier Series

Fourier Theorem: Every periodic function can be represented by the infinite series:

$$f(t) = a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \dots + b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \dots =$$
$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Fourier Analysis $\rightarrow$ find coefficients a_0 , a_n and b_n

Useful integrals:

$$\int_0^T \sin(n\omega_0 t) dt = 0$$

$$\int_0^T \cos(n\omega_0 t) dt = 0$$

$$\int_0^T \sin(k\omega_0 t) \cos(n\omega_0 t) dt = 0$$

$$\int_0^T \sin(k\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} 0, & k \neq n \\ \frac{T}{2}, & k = n \end{cases}$$
$$\int_0^T \cos(k\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} 0, & k \neq n \\ \frac{T}{2}, & k = n \end{cases}$$

Find coefficients...

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

We integrate over the period T

$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

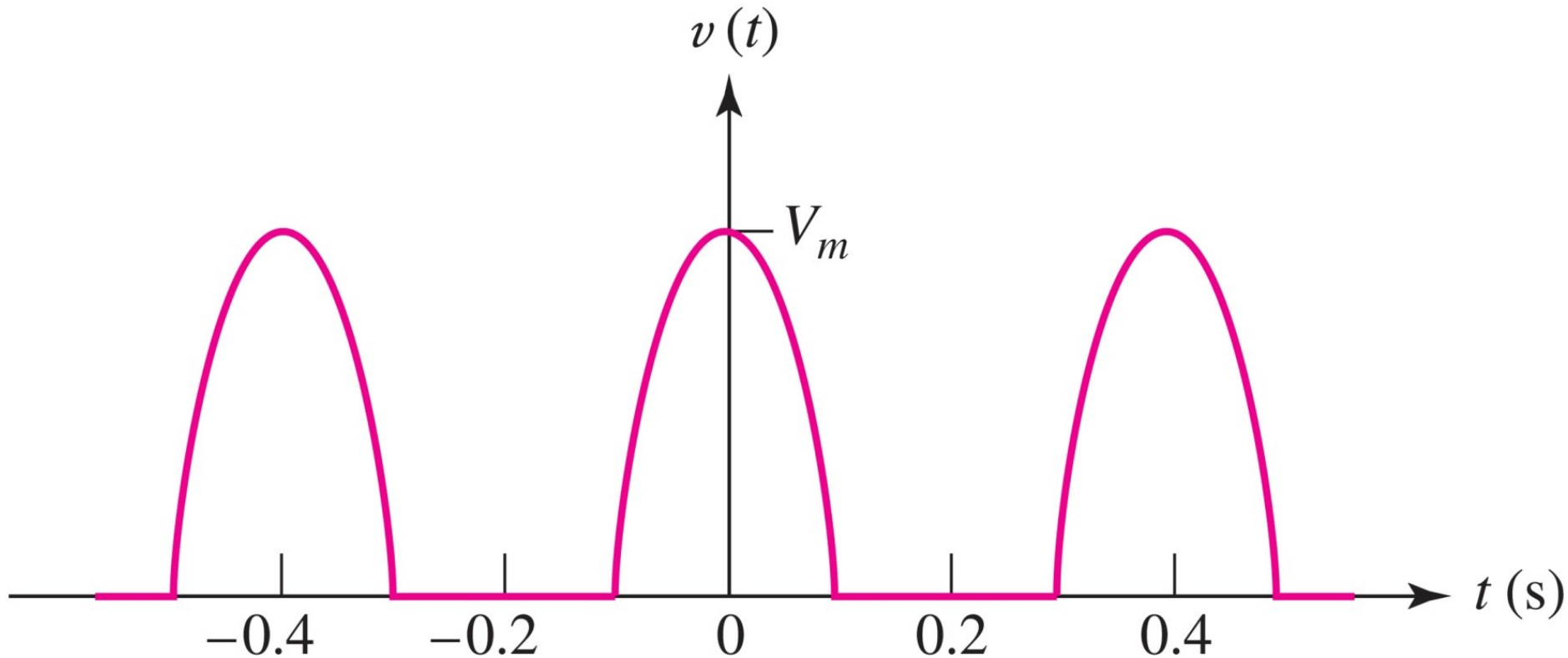
Integrate over the period T and multiply by $\cos(k\omega_0 t)$ or $\sin(k\omega_0 t)$ – **Fourier trick**

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt$$

Find the Fourier series representation of the waveform below:

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$$\sin^2 \frac{\alpha}{2} = \frac{1 - \cos \alpha}{2}$$

$$\cos^2 \frac{\alpha}{2} = \frac{1 + \cos \alpha}{2}$$

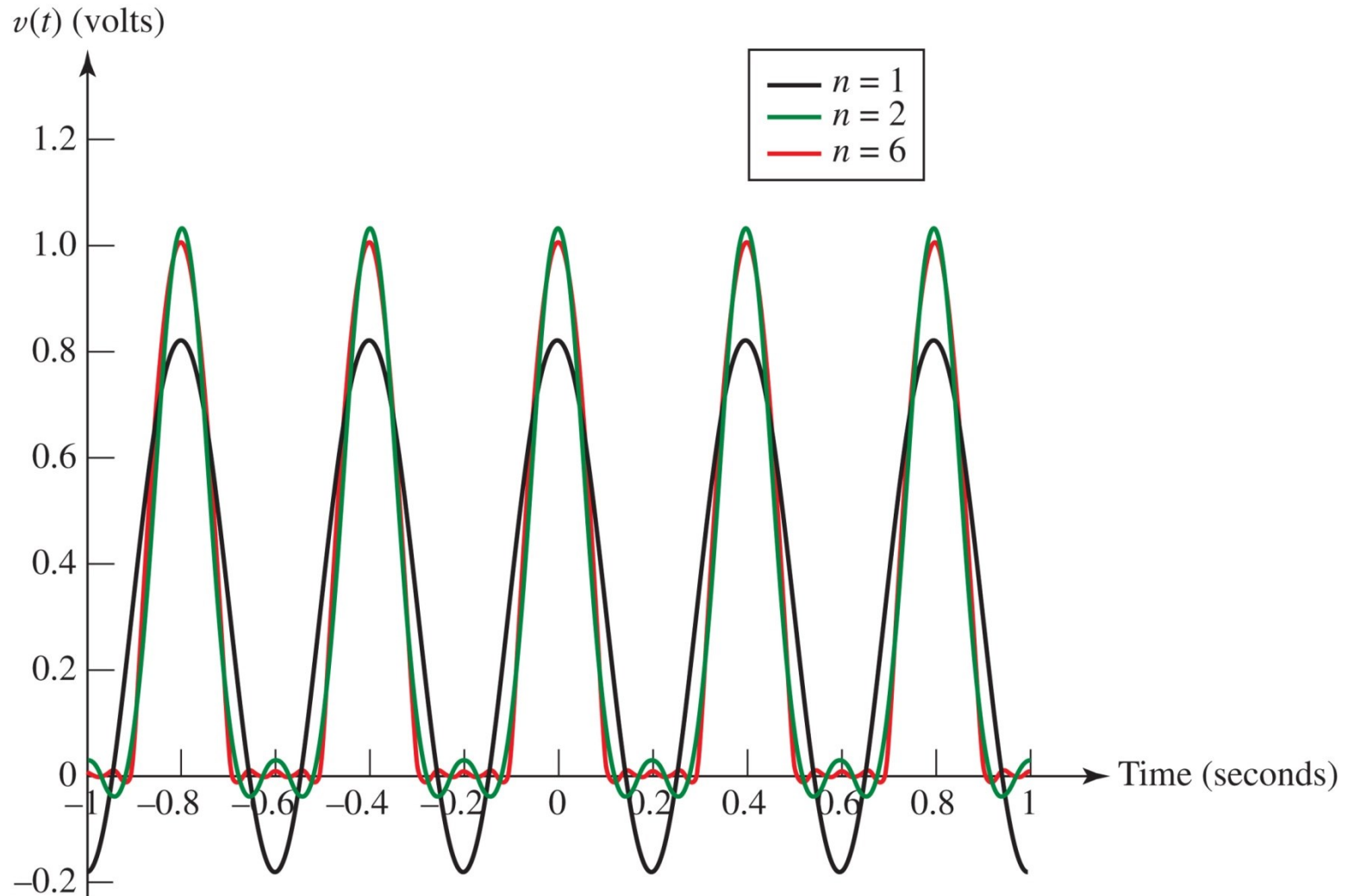
$$\sin \alpha \pm \sin \beta = 2 \sin \frac{\alpha \pm \beta}{2} \cos \frac{\alpha \mp \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = 2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

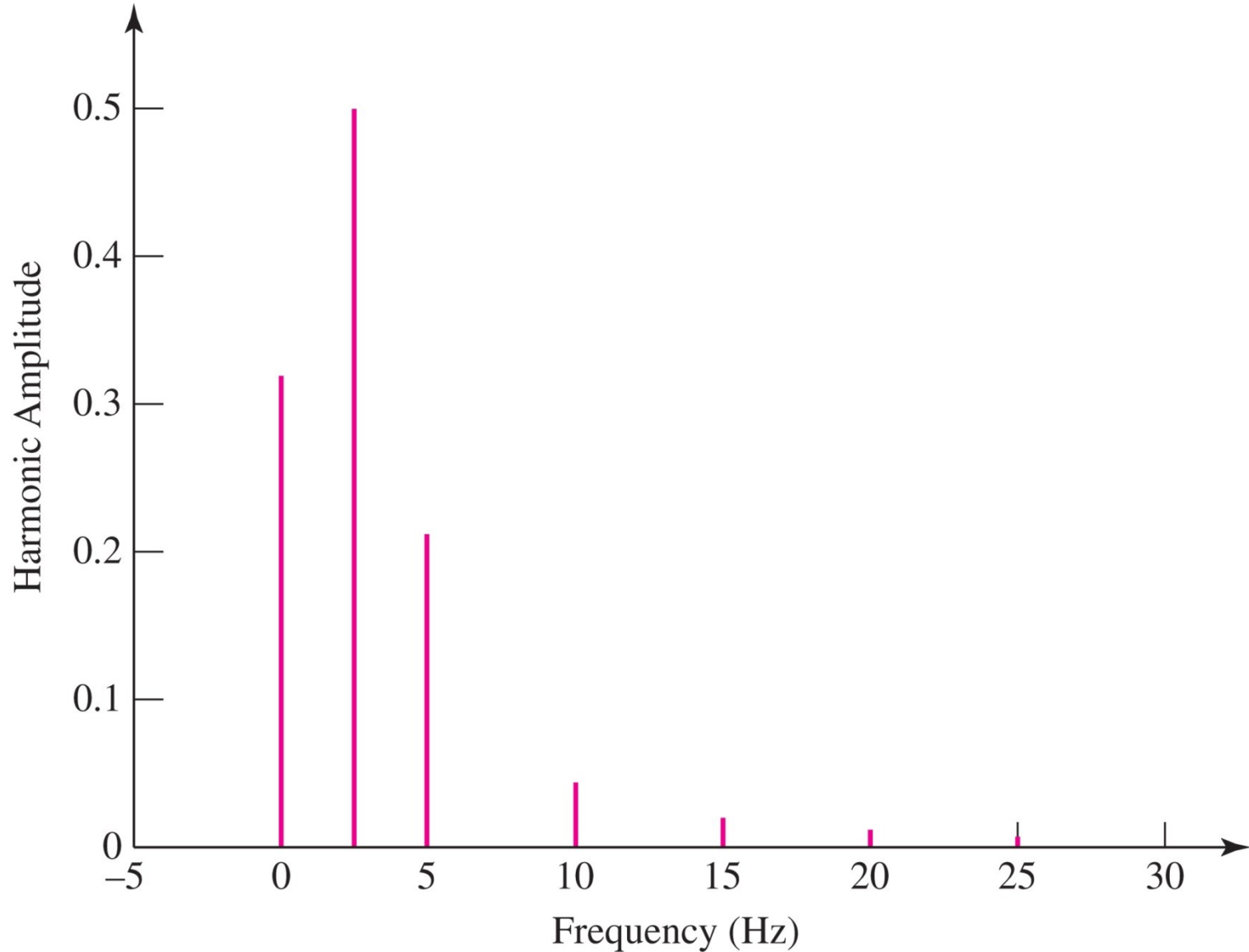
Fourier series representation of the above waveform in ***time domain*** would look like:

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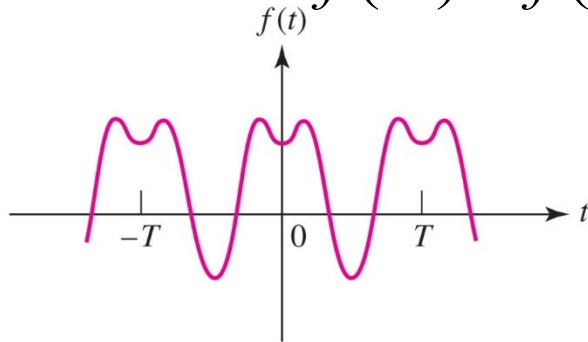
Can be represented in ***frequency domain*** using ***line (or discrete) spectrum*** :

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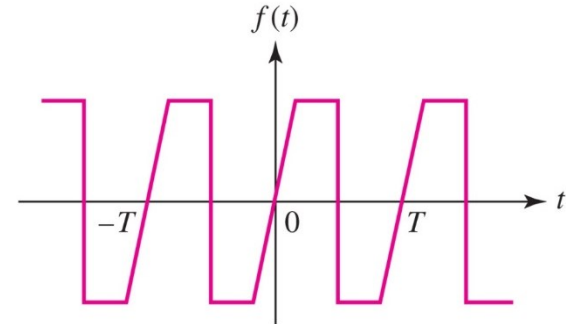


The use of symmetry

Even function $f(-t) = f(t)$



Odd function $f(-t) = -f(t)$



Product of any **two even** or any **two odd** functions → **even** function

Product of **even** and **odd** functions → **odd** function

Sum of **even** functions → **even** function

Sum of **odd** functions → **odd** function



No sum of odd functions can produce any even function other than 0!

So, for even $f(t)$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(n\omega_0 t) dt = 0$$

$$a_n = \frac{4}{T} \int_0^{T/2} f(t) \cos(n\omega_0 t) dt$$

So, for odd $f(t)$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t) dt = 0$$

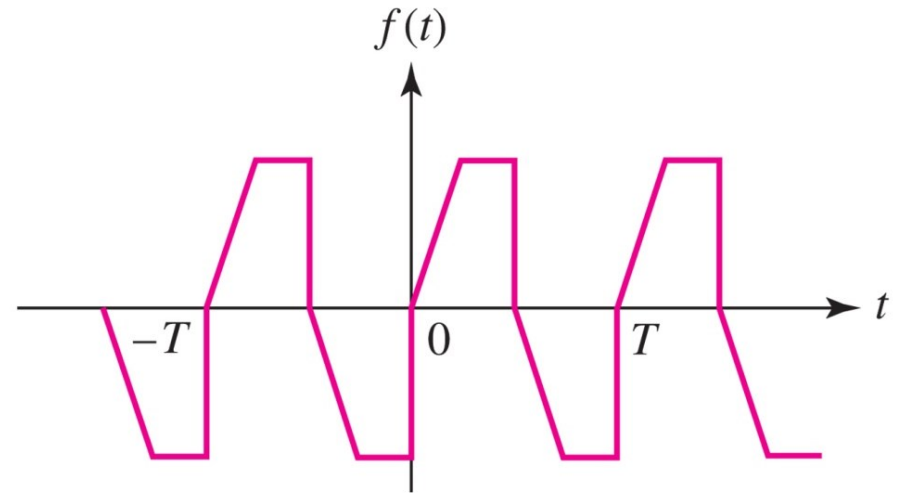
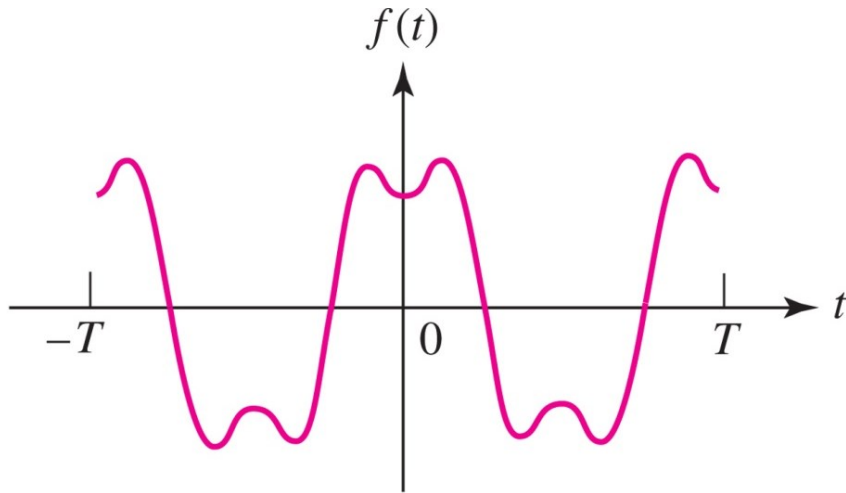
$$a_0 = 0$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt$$

Half-Wave Symmetry

$f(t)$ possesses half-wave symmetry if

$$f\left(t - \frac{T}{2}\right) = -f(t) \quad \text{or} \quad f\left(t + \frac{T}{2}\right) = -f(t)$$



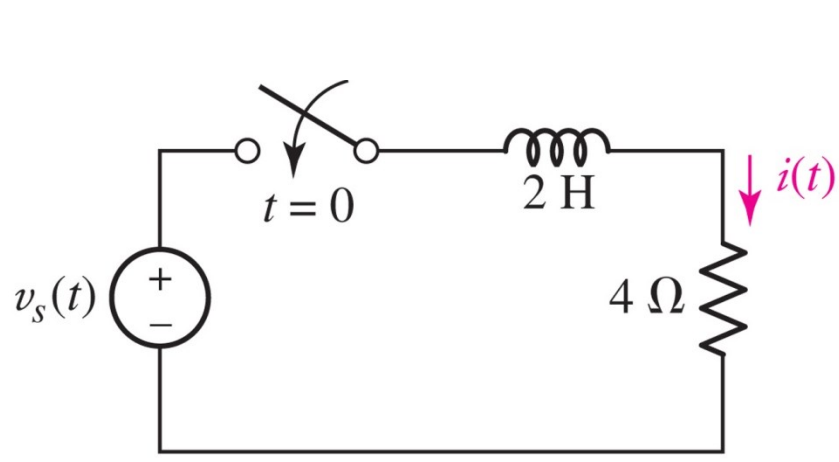
So, additionally

$$a_n = \begin{cases} \frac{4}{T} \int_0^{T/2} f(t) \cos(n\omega_0 t) dt & \text{for odd } n \\ 0 & \text{for even } n \end{cases} \quad b_n = \begin{cases} \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt & \text{for odd } n \\ 0 & \text{for even } n \end{cases}$$

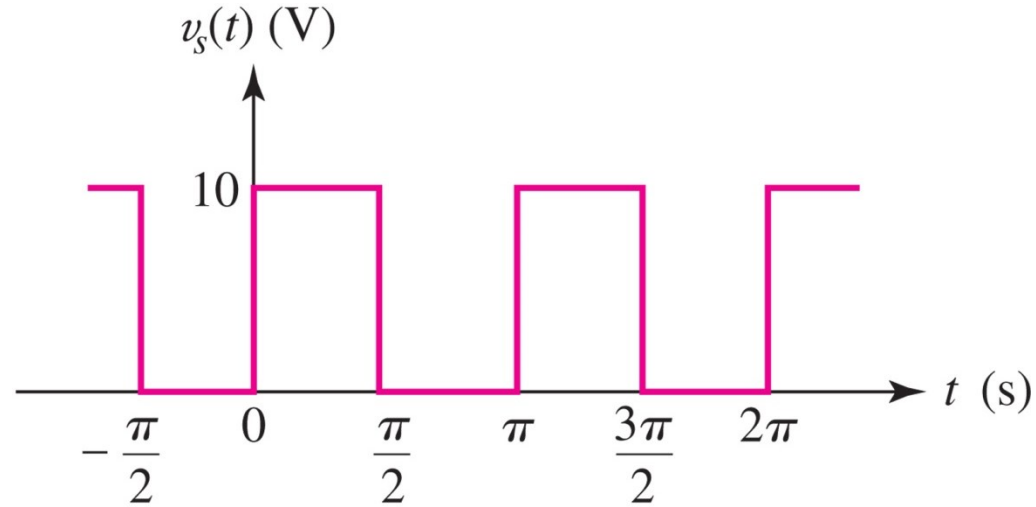
TABLE 18.1 Summary of Symmetry-Based Simplifications in Fourier Series

Symmetry Type	Characteristic	Simplification
Even	$f(t) = f(-t)$	$b_n = 0$
Odd	$f(t) = -f(-t)$	$a_n = 0$
Half-Wave	$f(t) = -f\left(t - \frac{T}{2}\right)$ or $f(t) = -f\left(t + \frac{T}{2}\right)$	$a_n = \begin{cases} \frac{4}{T} \int_0^{T/2} f(t) \cos n\omega_0 t \, dt & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$ $b_n = \begin{cases} \frac{4}{T} \int_0^{T/2} f(t) \sin n\omega_0 t \, dt & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$
Half-Wave and Even	$f(t) = -f\left(t - \frac{T}{2}\right)$ and $f(t) = f(t)$ or $f(t) = -f\left(t + \frac{T}{2}\right)$ and $f(t) = f(t)$	$a_n = \begin{cases} \frac{8}{T} \int_0^{T/4} f(t) \cos n\omega_0 t \, dt & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$ $b_n = 0 \quad \text{all } n$
Half-Wave and Odd	$f(t) = -f\left(t - \frac{T}{2}\right)$ and $f(t) = -f(-t)$ or $f(t) = -f\left(t + \frac{T}{2}\right)$ and $f(t) = -f(-t)$	$a_n = 0 \quad \text{all } n$ $b_n = \begin{cases} \frac{8}{T} \int_0^{T/4} f(t) \sin n\omega_0 t \, dt & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$

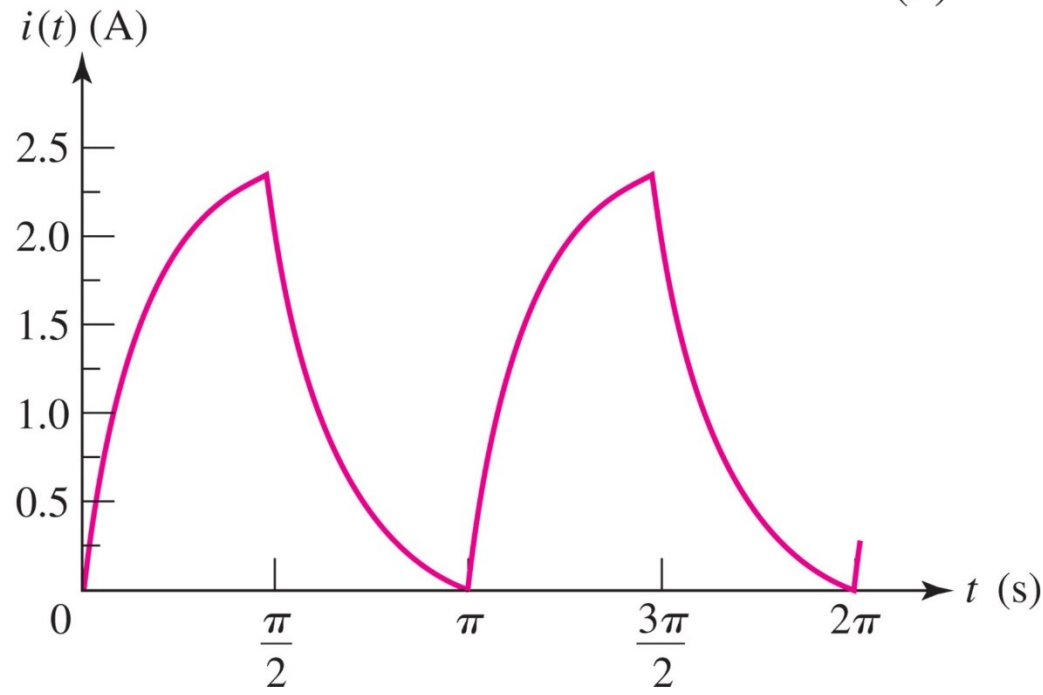
For the circuit in (a), determine the periodic response $i(t)$ corresponding to the forcing function shown in (b) if $i(0)=0$.



(a)



(b)



Complex form

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(e^{jn\omega_0 t} \frac{a_n - jb_n}{2} + e^{-jn\omega_0 t} \frac{a_n + jb_n}{2} \right)$$



$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt$$

Fourier Transform

Like in the case of Laplace transform, we have a transfer pair relationship

$$\mathbf{F(s)} = \int_{-\infty}^{+\infty} e^{-st} f(t) dt$$

Laplace transform

$$f(t) = \frac{1}{2\pi j} \int_{\sigma_0 - j\infty}^{\sigma_0 + j\infty} e^{st} \mathbf{F(s)} ds$$

Inverse Laplace transform

$$F(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega t} f(t) dt$$

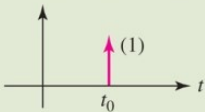
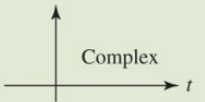
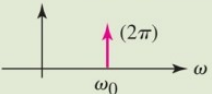
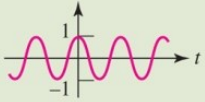
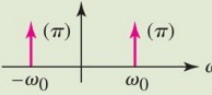
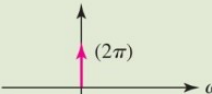
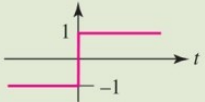
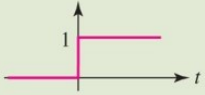
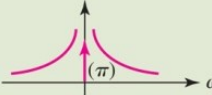
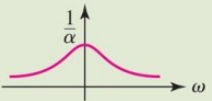
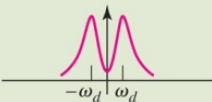
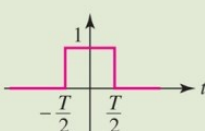
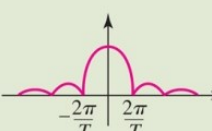
Fourier transform

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} F(j\omega) d\omega$$

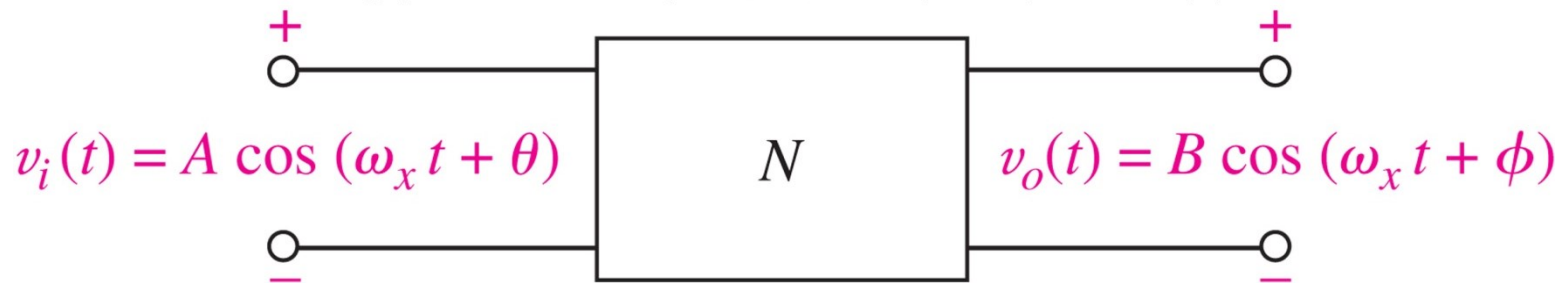
Inverse Fourier transform

The **Fourier transform** does not really care on the changing magnitudes of a signal, whereas the **Laplace transform** 'cares' both the changing magnitudes (exponential) and the oscillation (sinusoidal) parts. We can say that **Fourier transform** is a subset of **Laplace transform**.

TABLE 18.2 A Summary of Some Fourier Transform Pairs

$f(t)$	$f(t)$	$\mathcal{F}\{f(t)\} = F(j\omega)$	$ F(j\omega) $
	$\delta(t - t_0)$	$e^{-j\omega t_0}$	
	$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	
	$\cos \omega_0 t$	$\pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$	
	1	$2\pi \delta(\omega)$	
	$\text{sgn}(t)$	$\frac{2}{j\omega}$	
	$u(t)$	$\pi \delta(\omega) + \frac{1}{j\omega}$	
	$e^{-\alpha t} u(t)$	$\frac{1}{\alpha + j\omega}$	
	$[e^{-\alpha t} \cos \omega_d t] u(t)$	$\frac{\alpha + j\omega}{(\alpha + j\omega)^2 + \omega_d^2}$	
	$u(t + \frac{1}{2}T) - u(t - \frac{1}{2}T)$	$T \frac{\sin \frac{\omega T}{2}}{\frac{\omega T}{2}}$	

System Function and response in frequency domain

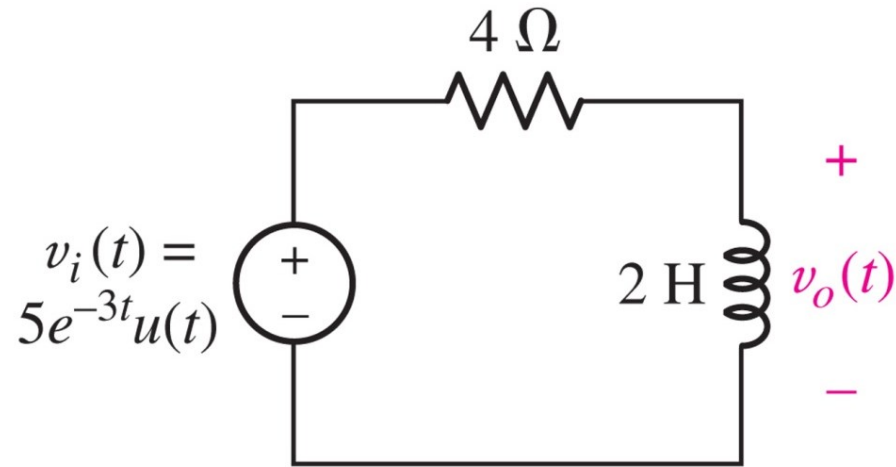


Transfer (system) function

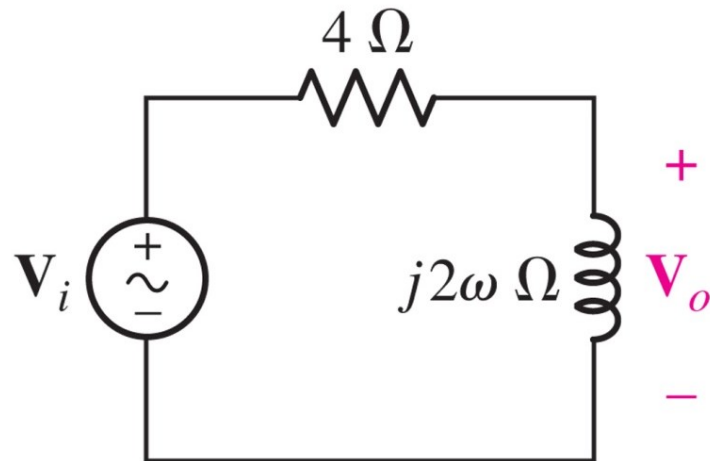
$$\mathbf{H}(j\omega) = \frac{\mathbf{F}(v_i(t))}{\mathbf{F}(v_o(t))}$$

Contrary to Laplace transform, Fourier transform enables to handle nonsinusoidal forcing functions and responses

Find the voltage across the inductor of the circuit shown when the input voltage is a simple exponentially decaying pulse as indicated.

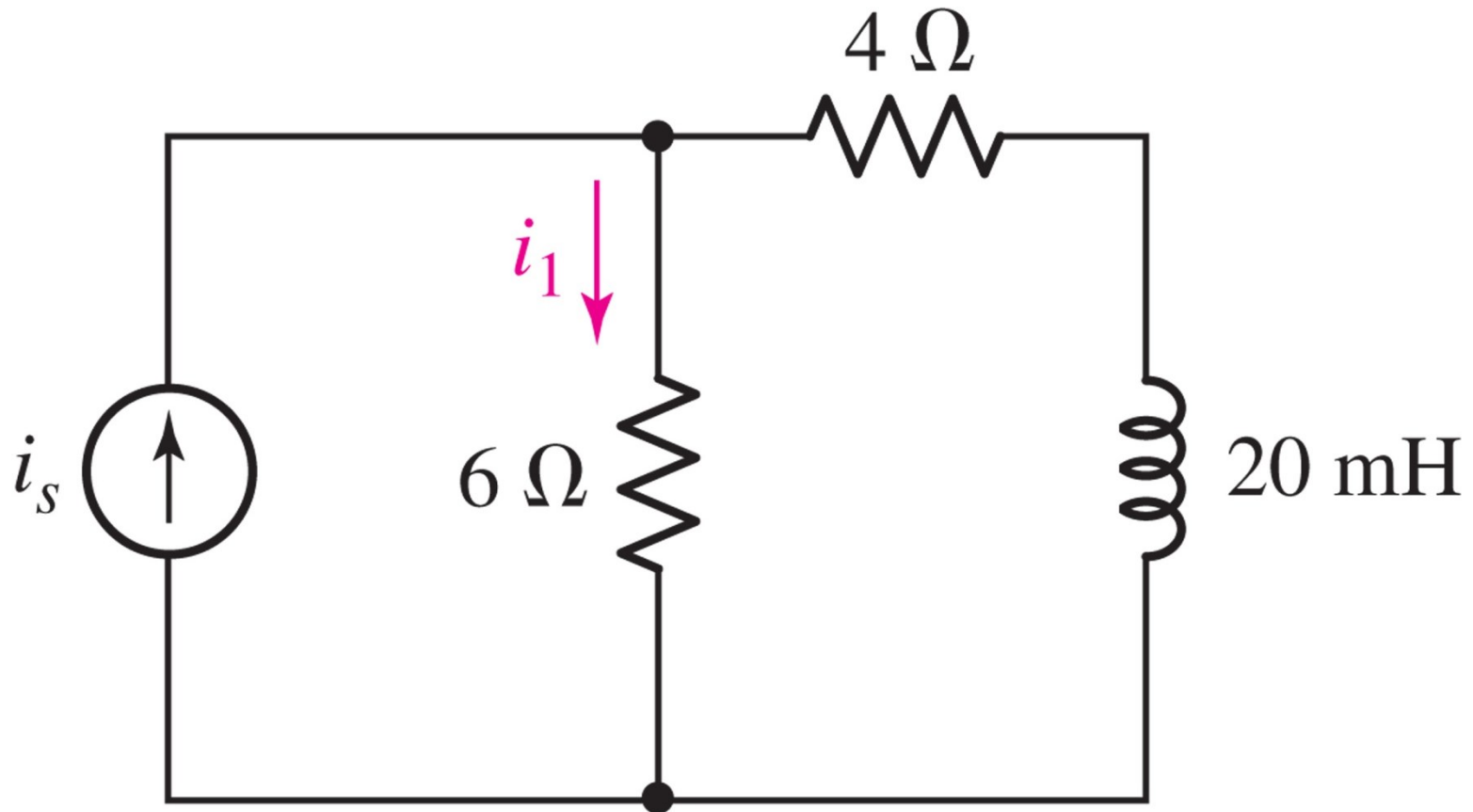


(a)



(b)

Use Fourier transform techniques to find $i_1(t)$ at $t=1.5$ ms if i_s equals (a) $u(t)$ A; (b) $\cos 500t$ A



a) 0.683 A; b) 0.308 A